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effective of the evaluated prediction models. The study combines physical measurement,

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statistical validation, and data-driven prediction in one workflow. The findings are intended to support early-stage configuration selection; further validation with controlled mechanical input, electrical load measurements, and independent experimental data is needed before broader practical application.

BUYSURE AI: An AI-Based Platform for Smarter Online Shopping Decisions

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Abstract

Online shopping has become an essential part of modern life, providing users with a wide range of products and convenient purchasing options. However, selecting the right product remains challenging due to fake reviews, misleading product information, fluctuating prices, unreliable sellers, and the large number of similar products available online. Users often depend on ratings and reviews to make purchasing decisions, but these factors alone may not provide an accurate understanding of a product's actual quality and value. BUYSURE AI is an AI-based shopping assistant developed to help users make safer, smarter, and more informed purchasing decisions. The proposed system uses Machine Learning (ML) and Natural Language Processing (NLP) techniques to analyze product details, customer reviews, ratings, prices, features, and comparable products. The system identifies suspicious or potentially misleading reviews and analyzes genuine customer feedback to extract important positive and negative aspects of a product. It also compares multiple products based on price, features, quality, ratings, and user-defined requirements. Based on the collected and analyzed information, BUYSURE AI generates a simple and understandable recommendation such as "Buy," "Wait," or "Avoid," along with the key reasons supporting the decision. Users can provide their budget, preferences, and specific requirements to receive personalized recommendations. Future enhancements may include product image and screenshot analysis, price tracking, seller risk detection, and automatic identification of better alternatives. The primary objective of BUYSURE AI is to reduce the complexity of online shopping by transforming large amounts of product and review data into meaningful insights, enabling users to make reliable and informed purchasing decisions.

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Keywords— BUYSURE AI, online shopping, machine learning, natural language processing, review analysis, product comparison, personalized recommendations.